Homework 3 - Rhema Phiri

1. EC2 setup description

The analysis was performed on an Amazon EC2 instance running Amazon Linux 2023. PySpark was installed in an Anaconda Python 3.11 environment. Spark was initialized in local mode, utilizing all available CPU cores on the single instance. Data was downloaded from the course S3 bucket to local disk using the AWS CLI before being loaded into Spark.

2. Dataset Month Used

Both the Yellow Taxi and Green Taxi trip record datasets for January 2026 were used, sourced from the NYC Taxi & Limousine Commission (TLC) and stored in the course S3 bucket as Parquet files. The datasets were standardized and combined, ensuring that all columns from both sources are preserved.

3. Data Cleaning Assumptions

- Removing null pickup or dropoff timestamps was carried out because trips without times cannot be analyzed temporally
- Removing negative trip distance was done because this indicates an invalid or cancelled trip
- Removing rows with a negative fare amount was necessary as negative fares are billing errors and are invalid trips. Similarly, negative total amount are likely data entry or system errors
- Removing rows with trip duration over 24 hours was carried out as this does not make sense and has to be a data error
- Lastly, removing rows with dropoff time before pickup time was essential as this is physically impossible and data contains corrupt timestamps

4. Analytics

Q1: Which taxi type had more trips?

Taxi Type	Count
Yellow	3518128
Green	38909

Yellow taxis had more trips than green taxis. This is because green taxis operate primarily in outer boroughs and are far less utilized than yellow taxis in January 2026.

Q2: Average fare by taxi type?

Taxi Type	Average Fare
Yellow	$21.08
Green	$16.00

Yellow taxis charge more on average. When context is taken into consideration, this makes sense as it reflects more airport and Manhattan trips with higher base rates and longer durations.

Q3: What was the average trip distance by taxi type?

Taxi Type	Average Trip Distance
Yellow	6.76 miles
Green	12.99 miles

Green taxis travel nearly twice the average distance of yellow taxis. This is consistent with green taxis serving outer boroughs where trips tend to cover longer distances

Q5: What percentage of trips were under 2 miles?

51.95% of trips were under 2 miles. This reveals that the majority of NYC taxi trips are short-distance rides.

Q8: Fare Prediction Model using Random Forest

A Random Forest Regressor was trained using PySpark MLlib to predict fare amount from non-fare trip features. An 80/20 split was used to create a training and testing data subset. Feature columns used include `trip_distance, PULocationID, DOLocationID, passenger_count, RatecodeID, payment_type, congestion_surcharge, taxi_type.`

The table below shows evaluation results:

Metric	Result
Training RMSE	5.8434
Testing RMSE	6.1287
Testing MAE	2.8301
Testing R^2	0.8895

The model achieves an R-Squared of 0.89 on the test set, explaining 89% of fare variance. The small gap between the training RMSE (5.84) and testing RMSE (6.13) indicates the model generalizes well without significant overfitting.

Figure 1 below shows this. The model performs well for typical mid-range trips but struggles with outliers eg. very cheap short trips and very expensive long-distance trips. This is common for random forests, which tend to predict toward the mean when unsure.

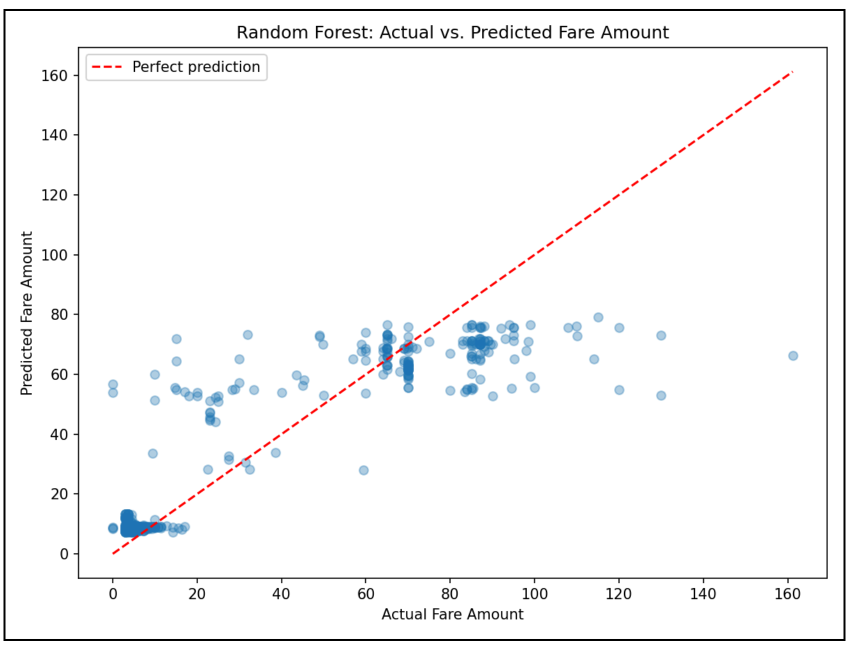

Figure 1: Random Forest - Actual vs Predicted Fare Amounts

Figure 2 below shows the importance of various predictors:

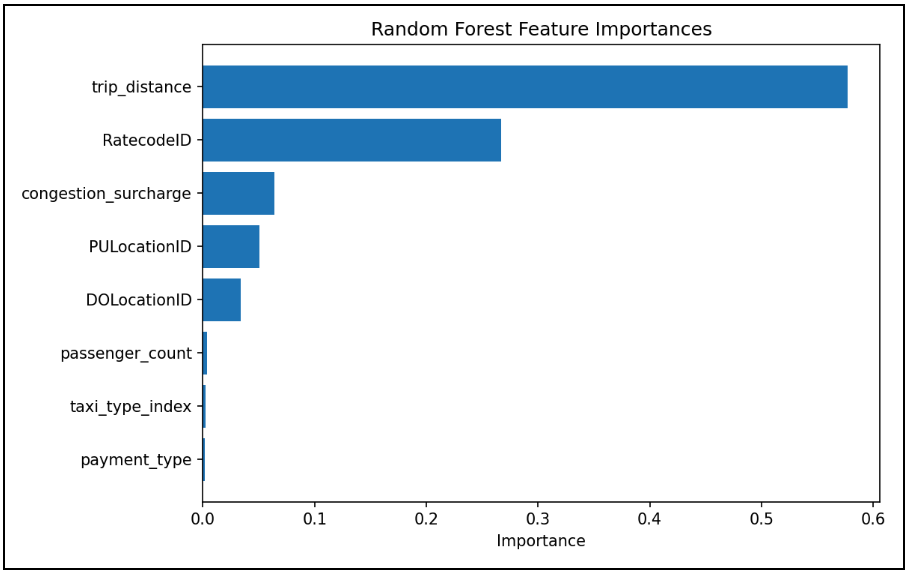

Figure 2: Random Forest Feature Importances

`trip_distance` and `RatecodeID` were the most important predictors, as expected as distance directly drives the meter, and the rate code captures airport/special-rate trips. Location IDs contributed moderately which reflects fare zone differences across the city.